

Ethan Chang

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OBJECTIVE

Computer Science B.S. student (May 2027) seeking full-time Software Engineering, Data Engineering, Data Science, or AI roles.

EDUCATION

Rochester Institute of Technology

Bachelor of Science, Computer Science; GPA: 3.6/4.0

Dean's List: Fall 2024, Spring 2025, Fall 2025, Spring 2026

Expected May 2027

Rochester, NY

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, C, C#, R, HTML/CSS

Frameworks & Libraries: React, Node.js, Express, Angular, Spring Boot, Pandas, NumPy, scikit-learn, XGBoost, Matplotlib, JUnit

AI & LLMs: Claude API, DeepSeek-R1, Prompt Engineering, Text Embeddings

Tools & Platforms: Git, GitHub Actions, Docker, PostgreSQL, Linux, Maven, Jupyter Notebook, DataGrip, Vercel, Render

EXPERIENCE

Sandia National Laboratories

Data Science R&D Intern

May 2026 – Present

Albuquerque, NM (Onsite)

- Built a training tool that simulates data from hypothetical radioactive material releases, partnering with laboratory scientists to translate nuclear-science data into software features.
- Earned a **Thunderbird Award** (Purpose-Driven category) for contributions during the internship.

Artificial Intelligence and Biomedical Informatics and Data Science (AIBIDS) Program

Data Science Intern

Jun 2025 – Aug 2025

Pittsburgh, PA (Hybrid)

- Surfaced risk patterns in **5,000+** de-identified patient records to inform a predictive model for unplanned surgery.
- Extracted **464** pre-surgery clinical notes from a UPMC ACLR database (**2013–2021**) using SQL CTEs, joins, and embeddings.
- Classified activity levels with an LLM (DeepSeek-R1 14B), reaching **0.89** F1 on elite athletes via **6** hand-annotated example batches.
- Built logistic regression models in scikit-learn to predict unplanned surgery from social determinants of health (SDoH).
- Presented a research poster on activity level as an SDoH predictor to faculty, peers, and stakeholders.

NTID Learning Center

Software Development Tutor

Sep 2025 – Present

Rochester, NY (Onsite)

- Build students' coding confidence by teaching debugging and computational thinking in 1:1 and group sessions.
- Make Python, Java, and Excel accessible to Deaf and hard-of-hearing students by teaching in ASL and spoken English.

PROJECTS

PlayIQ: Sports Analytics Dashboard | React, Node.js, Python, XGBoost, Claude API

Mar 2026 – Present

- Built a React dashboard for **6** leagues (MLB, NBA, WNBA, NFL, NHL, NCAAB) with per-game matchup, form, and roster tabs.
- Engineered Log5, Poisson, and Binomial projection models; automated weekly XGBoost money-line retraining via GitHub Actions.
- Served live ESPN and MLB data via a cached, dependency-free Node.js backend with Claude API game analysis.

Secure Distributed Messenger | C#, .NET, TCP/UDP Sockets, Cryptography

Jan 2026 – Apr 2026

- Built a peer-to-peer encrypted messenger with a team of **4**, using AES-256 keys exchanged via RSA-2048 and signed messages.
- Owned peer discovery and fault tolerance: UDP broadcast discovery, **15**-second heartbeat timeouts, and backoff reconnection.

Chroma: U-Fund Crowdfunding Platform | Angular, TypeScript, Java, Spring Boot

Aug 2025 – Dec 2025

- Built a full-stack charity crowdfunding platform with a team of **4**: **15** Angular components over a Spring Boot REST API.
- Designed a layered controller/model/DAO architecture with swappable persistence, covered by **15** JUnit test classes.

Movie Database Management System | Python, PostgreSQL, SQL, SSH

Jan 2025 – May 2025

- Built an ETL pipeline with a team of **5** that cleans Kaggle datasets and loads them into a normalized PostgreSQL schema.
- Developed a modular Python application with SSH tunneling for secure remote database access and management.

LEADERSHIP & INVOLVEMENT

RIT/NTID Cross-Registered Student Advisory Board (XSAB)

Golisano College of Computing and Information Sciences (GCCIS) Representative

Sep 2025 – May 2026

Rochester, NY

- Represented cross-registered students within the RIT/NTID community, sharing XSAB updates on events, resources, and services.
- Selected by the NTID Co-op and Career Center as NTID Career Fair Ambassador in 2025 and 2026.